CPU 16-8-16 Project Report

This report documents the design and verification of a microcoded 16-bit instruction, 8-bit register, 16-bit address CPU. The architecture was built using Synthagate tools, ModelSim/Questa for simulation, and ASMCreator for microprogram control logic development.

1. Design Overview

The CPU follows a 16-8-16 architecture:
- 16-bit instructions
- 8-bit general-purpose registers
- 16-bit addressing

The control logic is implemented using a microcoded control unit with reusable subASMs. Instructions are grouped into Initial ASMs and supported by micro-operations in the subASM library.

2. Toolchain

- ASMCreator: for creating instruction and microcode ASMs
- Synthagate: for constructing HLS and RTL designs
- ModelSim/Questa: for HLS and RTL simulation

3. Project Structure

The final project directories contain:
- Initial/: Instruction ASMs (lod, bun, bcz, etc.)
- subASM/: Micro-operation ASMs (Fetch1, AbsAdr, etc.)
- Components/: alu.vhd and memory VHDL files
- HLS/: Synthesized HLS design
- RTL/: RTL netlist and simulation setup

4. Instruction Set Overview

The CPU supports a basic set of operations including load/store, branching (conditional and unconditional), arithmetic/logic via AO instructions, and I/O. The instruction decoding is handled via microcode based on standard ASM and subASM design.

5. Microarchitecture

The ALU (`alu.vhd`) performs arithmetic and logical operations used by AO instructions. Memory models (`memAlt65536x8.vhd`, `memAlt65536x16.vhd`) support both data and instruction spaces.

6. HLS and RTL Simulation Results

Functional correctness of the CPU was validated through simulation. The same sequence of operations was tested on both HLS and RTL levels, and matching results were obtained.

Observed output values on `outr` signal:
- 0008: lod immediate
- 1234: lod/str long direct indexed
- 0002: bun long direct
- 0003: bun long direct indexed
- 0002: add short direct
- 0001: bcz long direct (zero 0)
- 0001: bcz long direct (zero 1)
- 0001: bcz long direct indexed (zero 0/1)

7. Conclusion

The 16-8-16 CPU was successfully implemented using a microcoded control strategy. It passed full verification through HLS and RTL simulation. The project demonstrated proper memory access, branching, conditional execution, and ALU functionality. Final output values match the expected behavior.

The following waveforms illustrate the `outr` signal behavior during simulation:

Figure 1: HLS Simulation Results

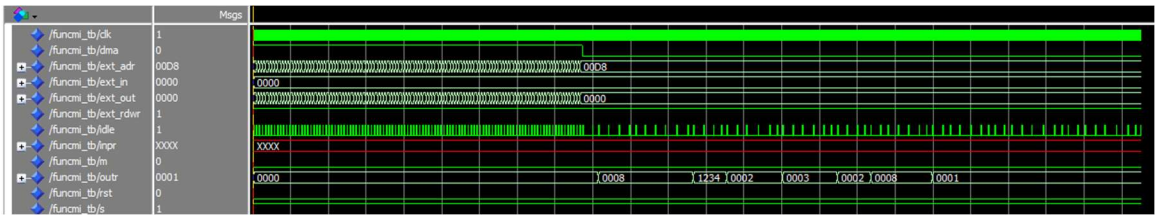
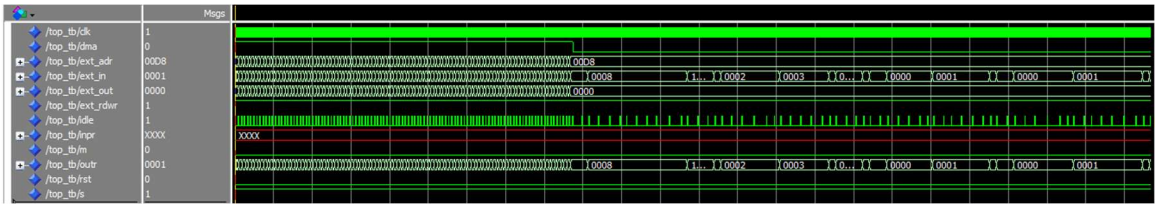


Figure 2: RTL Simulation Results


Msgs
/top_tb/clk
/top_tb/dma
/top_tb/ext_adr
/top_tb/ext_in
/top_tb/ext_out
/top_tb/ext_rdwr
/top_tb/idle
/top_tb/inpr
/top_tb/im
/top_tb/outr
/top_tb/rst
/top_tb/s
1
0
00D8
0001
0000
1
1
XXXX
0
0001
0
1
00D8
0008
1...
0002
0003
0...
0000
0001
0000
0001
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0008
1...
0002
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0...
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